



Lecture 4 - Knowledge Representation with Ontologies

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**FIZ Karlsruhe**

Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 4: Knowledge Representation with Ontologies

4.1 A Brief History of Ontologies

4.2 Why we do need Logic

Excursion 4: A Brief Recap of Essential Logics

Excursion 5: Description Logics

4.3 First Steps in OWL

4.4 More OWL

4.5 OWL and beyond

4.6 How to Design your own Ontology

The Semantic Web Technology Stack (not a piece of cake...)

Most apps use only a subset of the stack

Querying allows fine-grained data access

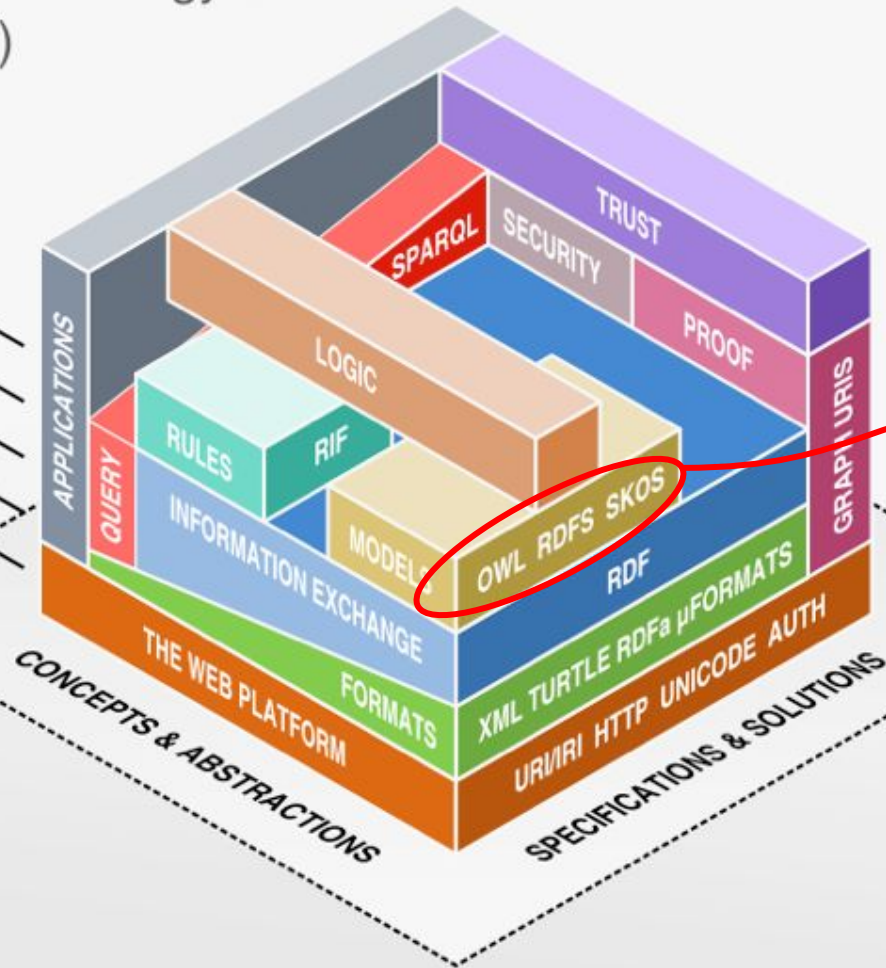
Standardized information exchange is key

Formats are necessary, but not too important

The Semantic Web is based on the Web

Linked Data uses a small
selection of technologies

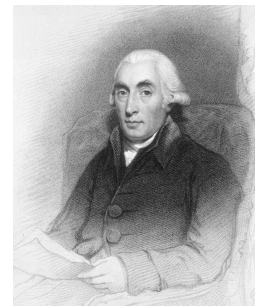
LINKED DATA



Web
Ontology
Language
(OWL)

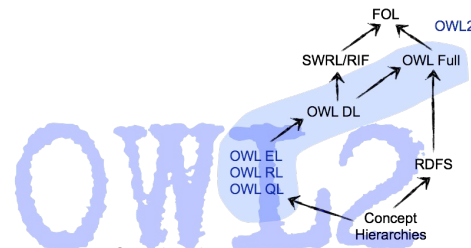
```
:Chemist a owl:Class .
:Physicist a owl:Class .
:JosephFourier a :Physicist .
:Jan_Baptist_van_Helmont a :Physicist .
:JosephBlack a Chemist .

:CarbonDioxideClub a owl:Class ;
    owl:oneOf
        ( :JosephFourier
          :Jan_Baptist_van_Helmont
          :JosephBlack ) .
```



```
CarbonDioxideClub ⊆ { JosephFourier,  
                      JanBaptistVanHelmont,  
                      JosepBlack }
```

- *There are only three scientists in the Carbon Dioxide Club.*



□

L

→

```

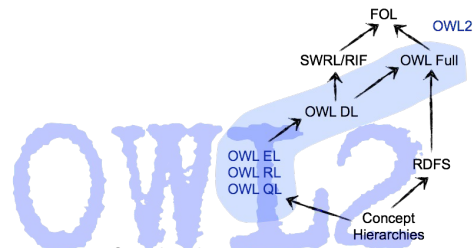
graph TD
    OWL2[OWL2] --> OWL_EL[OWL EL]
    OWL2 --> OWL_RL[OWL RL]
    OWL2 --> OWL_QL[OWL QL]
    OWL2 --> OWL_DL[OWL DL]
    OWL_DL --> SWRL_RIF[SWRL/RIF]
    OWL_DL --> OWL_Full[OWL Full]
    OWL_Full --> FOL[FOL]
    OWL_Full --> OWL2
    RDFS[RDFS] --> OWL2
    CH[Concept Hierarchies] --> OWL2
  
```

OWL Logical Class Constructors - Intersection

```
:Scientist a owl:Class .  
:ClimateActivist a owl:Class .  
:Scientists4Future a owl:Class ;  
    owl:intersectionOf (:Scientist :ClimateActivist) .
```

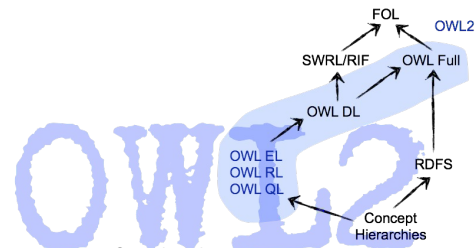
`Scientists4Future` $\equiv$ `Scientist` $\sqcap$ `ClimateActivist`

The class `:Scientists4Future` results from the intersection of all individuals of the classes `:Scientist` and `:ClimateActivist`



```
:Environmentalist a owl:Class ;
    owl:equivalentClass [
        owl:unionOf ( :ClimateActivist
                        :AnimalRightsActivist
                        :EnergySaver )
    ] .
```

Climate Activists, Animal Rights Activists, and Energy Savers are all Environmentalists

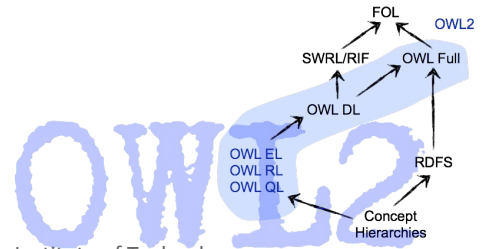


OWL Logical Class Constructors - Negation

```
:Pacifist a owl:Class .  
:Warmonger a owl:Class ;  
    owl:complementOf (:Pacifist) .
```

$\text{Warmonger} \equiv \neg \text{Pacifist}$

The class `:Warmonger` results from the complement of the class `:Pacifist`



OWL Property Restrictions

OWL property restrictions are used to describe **complex classes** via **properties**

restrictions on values:

`owl:hasValue`

`owl:allValuesFrom`

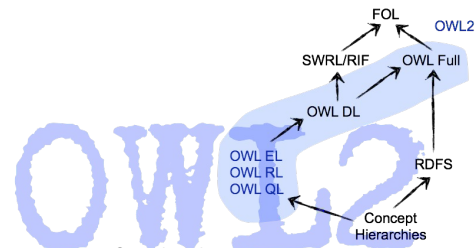
`owl:someValuesFrom`

restrictions on cardinality:

`owl:cardinality`

`owl:minCardinality`

`owl:maxCardinality`

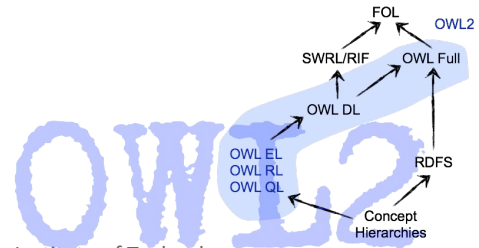


OWL Property Restrictions with Constants

```
:FouriersDiscoveries a owl:Class ;  
  rdfs:subClassOf  
    [ a owl:Restriction ;  
      owl:onProperty :discoverer ;  
      owl:hasValue :JosephFourier ] .
```

FouriersDiscoveries $\sqsubseteq$ discoverer.(JosephFourier)

The class **:FouriersDiscoveries** is described via fixed value assignment (=constant) of the individual **:JosephFourier** to the property **:discoverer**



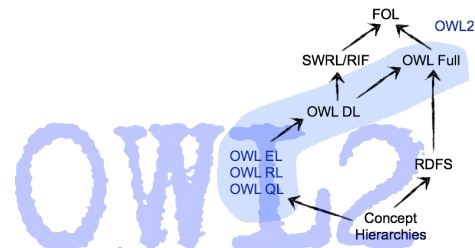
OWL Properties Restriction with Strict Binding

```
:VegetarianDish a owl:Class ;  
  rdfs:subClassOf  
    [ a owl:Restriction ;  
      owl:onProperty :ingredient ;  
      owl:allValuesFrom :VegetarianFood ] .
```

VegetarianDish $\sqsubseteq$
 $\forall \text{ingredient.VegetarianFood}$

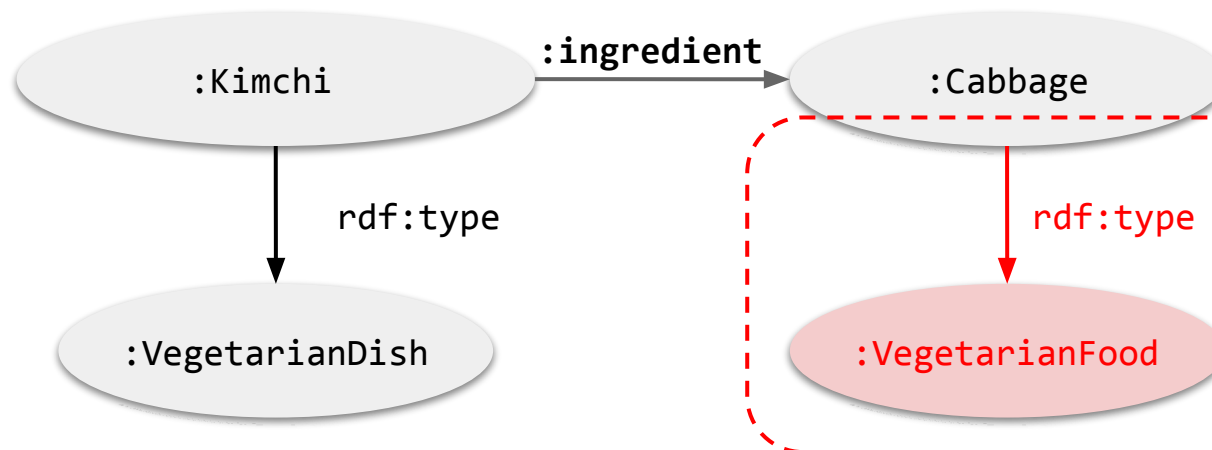
owl:allValuesFrom

fixes all instances of a specific class C
as allowed range for a property p
(strict binding) $\forall p.C$

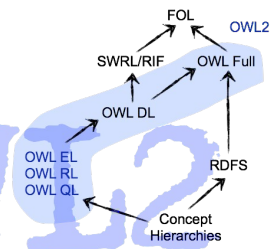


OWL Properties Restriction with Strict Binding

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:VegetarianDish a owl:Class ;
  rdfs:subClassOf
    [ a owl:Restriction ;
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```



Logical inference



OWL Property Restriction with Loose Binding

```

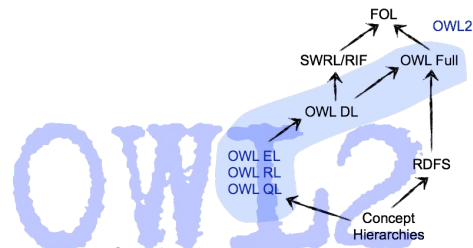
:Reader a owl:Class ;
  rdfs:subClassOf
    [ a owl:Restriction ;
      owl:onProperty :reads ;
      owl:someValuesFrom :Book ] .

```

Reader $\sqsubseteq \exists \text{reads}.\text{Book}$

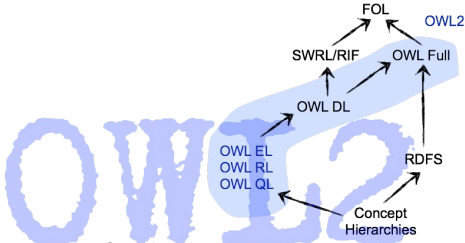
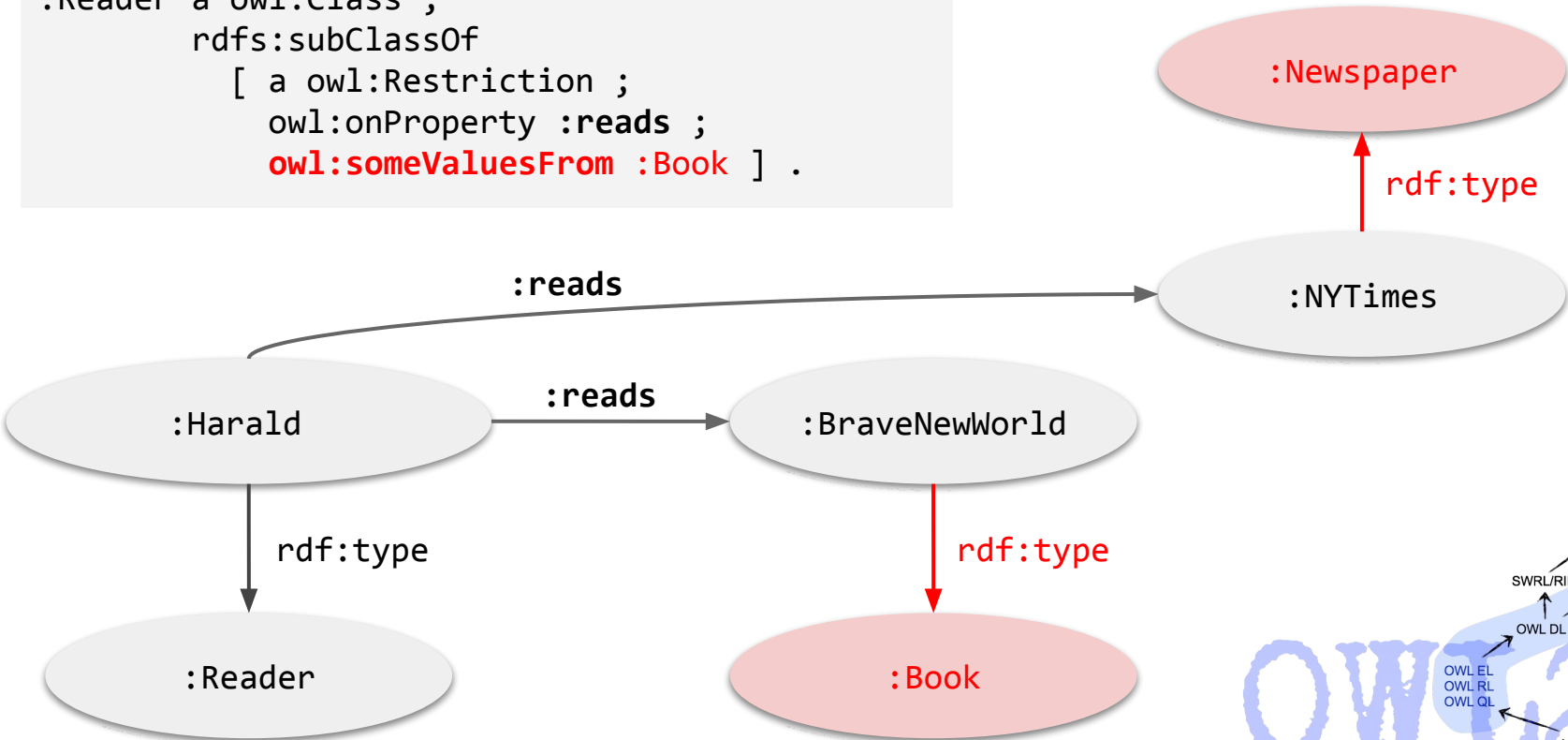
owl:someValuesFrom

describes that there must exist an individual for p
and fixes its range to class C (loose binding) $\exists p.C$



OWL Property Restriction with Loose Binding

```
:Reader a owl:Class ;  
  rdfs:subClassOf  
    [ a owl:Restriction ;  
      owl:onProperty :reads ;  
      owl:someValuesFrom :Book ] .
```



OWL Property Restrictions

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restrictions on values:

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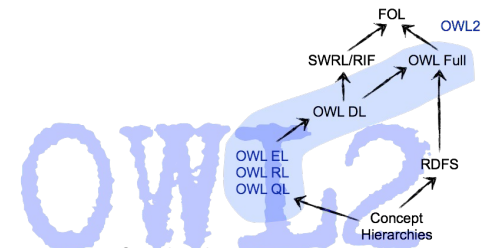
`owl:someValuesFrom`

restrictions on cardinality:

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`owl:minCardinality`

`owl:maxCardinality`



OWL Property Restrictions with Cardinality

```

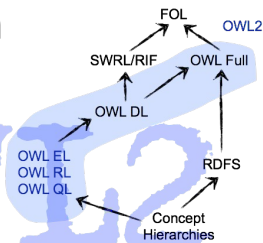
:StringQuartett a owl:Class ;
  rdfs:subClassOf
    [ a owl:Restriction ;
      owl:onProperty :member ;
      owl:cardinality "4"^^<http://www.w3.org/2001/XMLSchema#int> ] .

```

StringQuartett $\sqsubseteq$ =4.member.⊤

Class `:StringQuartett` is restricted to exactly 4 members, i.e. any instance of `:StringQuartett` must have exactly 4 values for the property `:member`

For `owl:maxCardinality` and `owl:minCardinality` the restriction gives upper and lower bounds on property value cardinalities.



A close-up photograph of an ancient coin, possibly a Greek drachma, featuring a detailed relief of an owl. The owl is depicted with large, circular eyes, a beak, and feathers. The coin is made of a dark, metallic material and shows signs of wear and damage, including cracks and a missing section on the right side. The text "OWL and Beyond" is overlaid in a bold, yellow, sans-serif font, slanted diagonally across the center of the image.

OWL and Beyond

Next Lecture...

Picture References:

- [1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [2]